Rishabh Jain

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PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years of experience bridging enterprise integration and production AI across GenAI, LLMs, RAG, NLP and computer vision. Built cloud deployed recommendation, document understanding and safety constrained robotics systems on AWS and GCP, engineering deterministic validation and guardrails around probabilistic models. MSc Robotics and Embedded AI candidate at Maynooth University, graduating August 2026; available from September 2026, Open to roles in India and Ireland, and to remote work globally. Indian citizen; full Irish work authorisation (Stamp 1G, no sponsorship required).

EDUCATION

Master of Science - Robotics and Embedded AI Sep 2025 – Exp. Aug 2026
Maynooth University Maynooth, Kildare, Ireland
Relevant Coursework: Deep Learning, AI & Language Processing, Computer Vision, HRI

Bachelor of Technology - Information Technology Aug 2016 – Oct 2020
Shri Vaishnav Vidyapeeth Vishwavidyalaya - SVIIT Indore, Madhya Pradesh, India

WORK EXPERIENCE

Maynooth University Jan 2026 – Present
Research Assistant Maynooth, Kildare, Ireland

Screen-based robot controls exclude many older adults and people with disabilities, while unconstrained LLM voice interfaces can turn a single misclassified command into unsafe physical motion.

- Designed VoiceBT, a safety-constrained voice-to-action framework that restricts a 2–4B parameter on-device Gemma model to intent classification while deterministic behaviour trees control sequencing and safety-critical execution.
- Achieved 100% desired-behaviour success and zero unsafe motions across a 43-case command corpus and 120 forced-misclassification trials in simulation, using explicit code-level safety guards to intercept unsafe actions.
- Integrated and tested the framework across multiple FarmBot systems at Maynooth University's farm using firmware-confirmed, tick-and-verify execution.

INFOSYS LIMITED Jul 2024 – Sep 2025
Senior Associate Consultant Indore, Madhya Pradesh, India

A loyalty programme showed every member the same broad-tier offers, missing revenue from personalisation but collaborative filtering alone cannot serve new members with no purchase history, and an LLM cannot sit in the critical path of a sub-second response at the pump.

- Led client discovery sessions and defined the AI architecture, integration requirements, and deployment strategy for an LLM-powered recommendation engine, part of a loyalty platform redesign at a major oil and gas company.
- Architected and delivered a hyperpersonalisation system on AWS Bedrock for LLM contextual ranking of customer offers, solving new member cold start with a hybrid engine that combines collaborative filtering and LLM ranking to lift new user engagement.
- Engineered real-time data ingestion pipelines using Dell Boomi on AWS EC2, connecting legacy POS transaction data to AWS SageMaker inference endpoints and achieving sub-second recommendation latency via an Amazon ElastiCache caching layer.

TATA CONSULTANCY SERVICES

Systems Engineer - AI Engineer Jan 2023 – Jul 2024
Mumbai, Maharashtra, India

Template based OCR broke whenever a supplier changed invoice layout, dumping large volumes into a manual review queue and driving an overdue invoice backlog that cost early payment discounts and strained supplier relationships.

- Developed a Python-based invoice document-understanding system using fine-tuned layout-aware transformer models, achieving 90%+ field-level extraction accuracy across varying invoice layouts and reducing manual processing time by 50%.
- Implemented a deterministic three-way-match gate across invoices, purchase orders, and goods receipts before payment routing, preventing incomplete, low-confidence, or inconsistent model outputs from entering the approval workflow.
- Deployed containerised inference services on GCP Cloud Run with Cloud Storage, Cloud Functions, Cloud Monitoring, and Cloud Logging, contributing to a 35% reduction in overdue invoices and improved SLA compliance.

Systems Engineer – DevOps Engineer Feb 2021 – Dec 2022

Mumbai, Maharashtra, India

Critical invoicing, payroll, warehousing, and inventory processes ran across disconnected legacy systems and data formats, with no reliable, monitored integration layer between them.

- Deployed and managed Dell Boomi runtime atoms on AWS EC2, configuring VPC networking, IAM roles, and CloudWatch alarms for production monitoring across invoicing, payroll, warehousing, and inventory pipelines.
- Implemented Dell Boomi and JavaScript middleware integrations across 10+ connector types (Salesforce, REST, SOAP, SFTP, HTTP, FTP, Disk) and formats (JSON, XML, Flat File) for a leading life sciences brand.
- Leveraged AWS S3 as a staging and archival layer for Disk/FTP connector workflows; provided real-time production support maintaining high-availability SLAs.

Earlier: Data Science Intern, TezMinds Software (Jan 2020 – Apr 2020) — DeepFake video classification with a custom CNN in TensorFlow, trained on 100,000+ videos.

TECHNICAL SKILLS

AI & Machine Learning — LLMs, RAG, AI Agents, NLP, Computer Vision, Transformers, Recommendation Systems, Model Evaluation, Guardrails

Cloud & Production AI — AWS Bedrock, SageMaker, EC2, S3, ElastiCache, GCP Cloud Run, Cloud Functions, Docker, Vector Databases, Model Deployment

Frameworks & Libraries — Python, PyTorch, TensorFlow, Hugging Face, LangChain, CrewAI, OpenCV, Scikit-learn, FastAPI

Integration & APIs — REST APIs, Dell Boomi, Enterprise Integration, Data Pipelines, JSON, XML, SOAP

PROJECTS

FlowGuard (Agentic Integration Incident Copilot)
Agentic AI · Enterprise Integration

- Developing an agentic incident response copilot that investigates API and enterprise data pipeline failures by correlating application logs, runbooks, API specifications, deployment records, dependency maps, and historical incidents.
- Designing a stateful investigation workflow for incident triage, evidence retrieval, dependency tracing, root cause synthesis, remediation planning, and human approval, with scoped tool access and auditable evidence linked recommendations.

DepthLens (Depth Aware Scene Descriptor)
Assistive Vision

- Built a real-time assistive-vision pipeline combining a 1–3B vision language model with Depth Anything V2 and YOLOv8n to generate scene descriptions enriched with object locations and relative depth.
- Fused detection and monocular depth outputs into structured spatial context before VLM inference, enabling the system to distinguish foreground, mid-ground, and distant objects while running on a consumer GPU.

Multilingual Discord Content Moderator
Reinforcement Learning · AI Safety

- Formulated multilingual content moderation as a Constrained Markov Decision Process across 13 languages, combining sentence embeddings, calibrated XLM-R toxicity scores, and user infraction history in a 403-dimensional observation space.
- Trained a graduated five-action policy Allow, Warn, Delete, Timeout, and Ban with hard action masking to prevent invalid or disproportionate enforcement decisions.

LANGUAGES

English — Fluent

Hindi — Native

CERTIFICATIONS

Generative-AI Cohort 1 by 100xEngineers

Generative AI with Large Language Models by DeepLearning.AI

Deep Learning Specialization by DeepLearning.AI